

Likelihood from Hamiltonian Flow: Derivation

Step 1: Initial Distribution

Stars are assumed to be drawn at $t = 0$ from a 2D Gaussian centred on the midplane:

$$f_0(z_0, w_0) = \frac{1}{2\pi \sigma_z \sigma_w} \exp\left(-\frac{z_0^2}{2\sigma_z^2} - \frac{w_0^2}{2\sigma_w^2}\right).$$

Step 2: Liouville's Theorem

Hamiltonian dynamics preserve phase-space volume: the Jacobian of any canonical flow Φ^t is exactly 1. Therefore the probability density at observation time t is obtained by *back-integrating* each observed point $(z_{\text{obs}}, w_{\text{obs}})$ to its initial condition $(z_0, w_0) = \Phi_{h_{dm}}^{-t}(z_{\text{obs}}, w_{\text{obs}})$:

$$f_t(z_{\text{obs}}, w_{\text{obs}}) = f_0\left(\underbrace{\Phi_{h_{dm}}^{-t}(z_{\text{obs}}, w_{\text{obs}})}_{(z_0, w_0) \text{ back-integrated}}\right).$$

No Jacobian factor appears — it cancels exactly.

Step 3: The Likelihood

For N independently observed stars:

$$\mathcal{L}(h_{dm}) = \prod_{i=1}^N \frac{1}{2\pi \sigma_z \sigma_w} \exp\left(-\frac{[z_0^{(i)}(h_{dm})]^2}{2\sigma_z^2} - \frac{[w_0^{(i)}(h_{dm})]^2}{2\sigma_w^2}\right)$$

Key point: $z_0^{(i)}$ and $w_0^{(i)}$ are the *back-integrated initial conditions*, not the observed values. h_{dm} enters exclusively through this back-integration.

Step 4: Log-Likelihood and Sensitivity

Taking the log:

$$\ln \mathcal{L}(h_{dm}) = \text{const} - \sum_{i=1}^N \left[\frac{[z_0^{(i)}(h_{dm})]^2}{2\sigma_z^2} + \frac{[w_0^{(i)}(h_{dm})]^2}{2\sigma_w^2} \right].$$

You maximise by finding h_{dm} such that back-integrated orbits are most consistent with starting near the midplane (small z_0 , small w_0).

The gradient with respect to h_{dm} is:

$$\frac{\partial \ln \mathcal{L}}{\partial h_{dm}} = - \sum_{i=1}^N \left[\frac{z_0^{(i)}}{\sigma_z^2} \frac{\partial z_0^{(i)}}{\partial h_{dm}} + \frac{w_0^{(i)}}{\sigma_w^2} \frac{\partial w_0^{(i)}}{\partial h_{dm}} \right].$$

The partial derivatives $\partial z_0 / \partial h_{dm}$ and $\partial w_0 / \partial h_{dm}$ are small because dark matter contributes only $\sim 8\%$ of the total gravitational force at typical orbit heights — the back-integrated orbit barely changes when h_{dm} is varied.

Step 5: Force Budget

The gravitational force is a sum over sech^2 components:

$$K_z(z) = -\frac{d\Phi}{dz} = -4\pi G \sum_{\alpha} \rho_{0,\alpha} h_{\alpha} \tanh\left(\frac{z}{h_{\alpha}}\right),$$

with parameters:

Component	$\rho_0 [M_{\odot}/\text{pc}^3]$	$h [\text{pc}]$	$\rho_0 h$	DM fraction at $z = 250 \text{ pc}$
Thin disk	0.04	200	8.0	
Thick disk	0.01	400	4.0	
Gas	0.05	800	40.0	
DM	0.01	250	2.5	8.3%
Total			54.5	

Gas alone ($\rho_0 h = 40$) is $16\times$ larger than dark matter ($\rho_0 h = 2.5$). DM never exceeds $\sim 9\%$ of the total force at any relevant height.

Step 6: Why the σ Mismatch Fails

With a wrong $\sigma_{z,\text{wrong}} < \sigma_{z,\text{true}}$, the biased log-likelihood is:

$$\ln \mathcal{L}_{\text{biased}}(h_{dm}) = \text{const} - \sum_i \left[\frac{[z_0^{(i)}]^2}{2\sigma_{z,\text{wrong}}^2} + \frac{[w_0^{(i)}]^2}{2\sigma_{w,\text{true}}^2} \right].$$

Shrinking σ_z *over-weights* the z_0^2 term, tilting the landscape. The shift of the peak scales as:

$$\Delta h_{dm} \propto \frac{\text{tilt from mismatch}}{\text{curvature of } \ln \mathcal{L}} \propto \frac{1/\sigma_{z,\text{wrong}}^2 - 1/\sigma_{z,\text{true}}^2}{\partial^2 \ln \mathcal{L} / \partial h_{dm}^2}.$$

Since the curvature (Fisher information in h_{dm}) is tiny due to the $\sim 8\%$ DM fraction, even a large mismatch in σ_z barely moves the optimum.